Chapter 13

Black Holes: A String/M-Theory Perspective

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he pre-string theory conflict between general relativity and quantum mechanics was an affront to our visceral sense that the laws of nature should fit together in a seamless, coherent whole. But this antagonism was more than a towering abstract disjunction. The extreme physical conditions that occurred at the moment of the big bang and that prevail within black holes cannot be understood without a quantum mechanical formulation of the gravitational force. With the discovery of string theory, we now have a hope of solving these deep mysteries. In this and the next chapter, we describe how far string theorists have gone toward understanding black holes and the origin of the universe.

Black Holes and Elementary Particles

At first sight it's hard to imagine any two things more radically different than black holes and elementary particles. We usually picture black holes as the most gargantuan of heavenly bodies, whereas elementary particles are the most minute specks of matter. But the research of a number of physicists during the late 1960s and early 1970s, including Demetrios Christodoulou, Werner Israel, Richard Price, Brandon Carter, Roy Kerr, David Robinson, Hawking, and Penrose, showed that black holes and elementary particles are perhaps not as different as one might think. These physicists found increasingly persuasive evidence for what John Wheeler has summarized by the statement "black holes have no hair." By this, Wheeler meant that except for a small number of distinguishing features, all black holes appear to be alike. The distinguishing features? One, of course, is the black hole's mass. What are the others? Research has revealed that they are the electric and certain other force charges a black hole can carry, as well as the rate at which it spins. And that's it. Any two black holes with the same mass, force charges, and spin are completely identical. Black holes do not have fancy "hairdos"—that is, other intrinsic traits—that distinguish one from another. This should ring a loud bell. Recall that it is precisely such properties—mass, force charges, and spin—that distinguish one elementary particle from